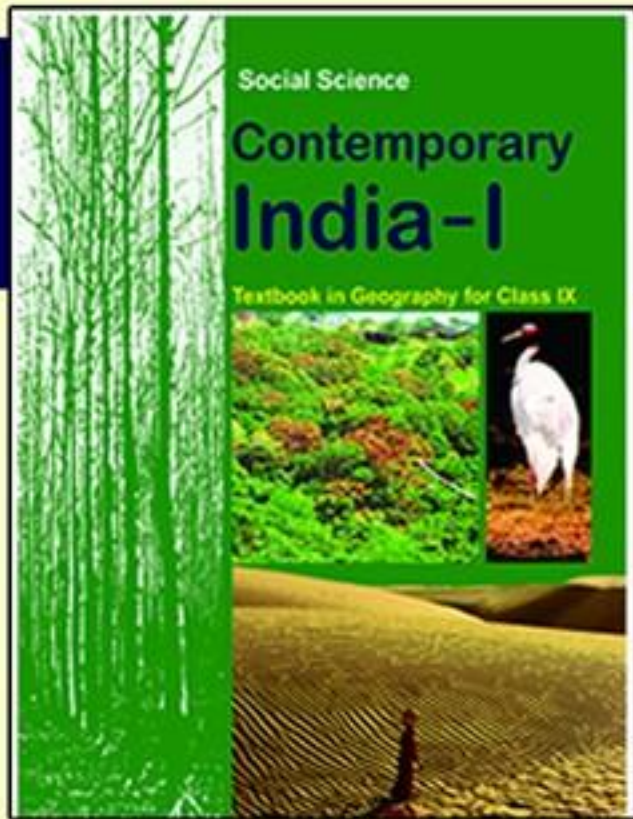




Learning **Space**



NCERT BASICS



GEOGRAPHY

- HEAT ZONES OF THE EARTH
- WINDS
- LATITUDES AND LONGITUDES PASSING THROUGH INDIA

MODULE - 4

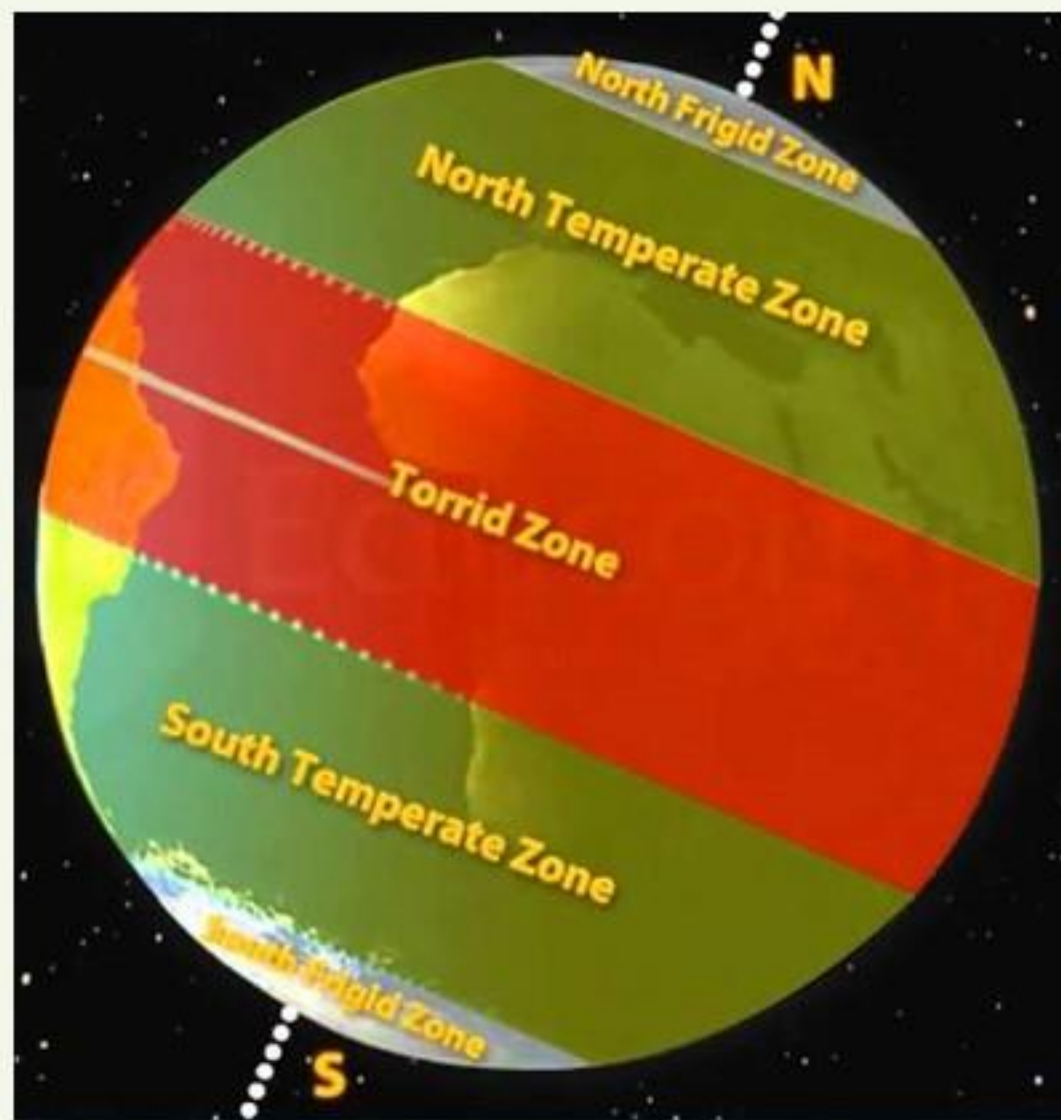




HEAT ZONES OF THE EARTH



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- There are 3 heat zones of the Earth:
- Torrid zone.
- Temperate zone.
- Frigid zone.



Learning Space



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A BRIEF ON HEAT ZONES

1. TORRID ZONE

- As the mid day Sun is exactly over head at least once a year on all the latitudes in between Tropic of Cancer and Tropic of Capricorn, this region receives the maximum heat.
- And it is called the Torrid Zone.

2. TEMPARATE ZONE

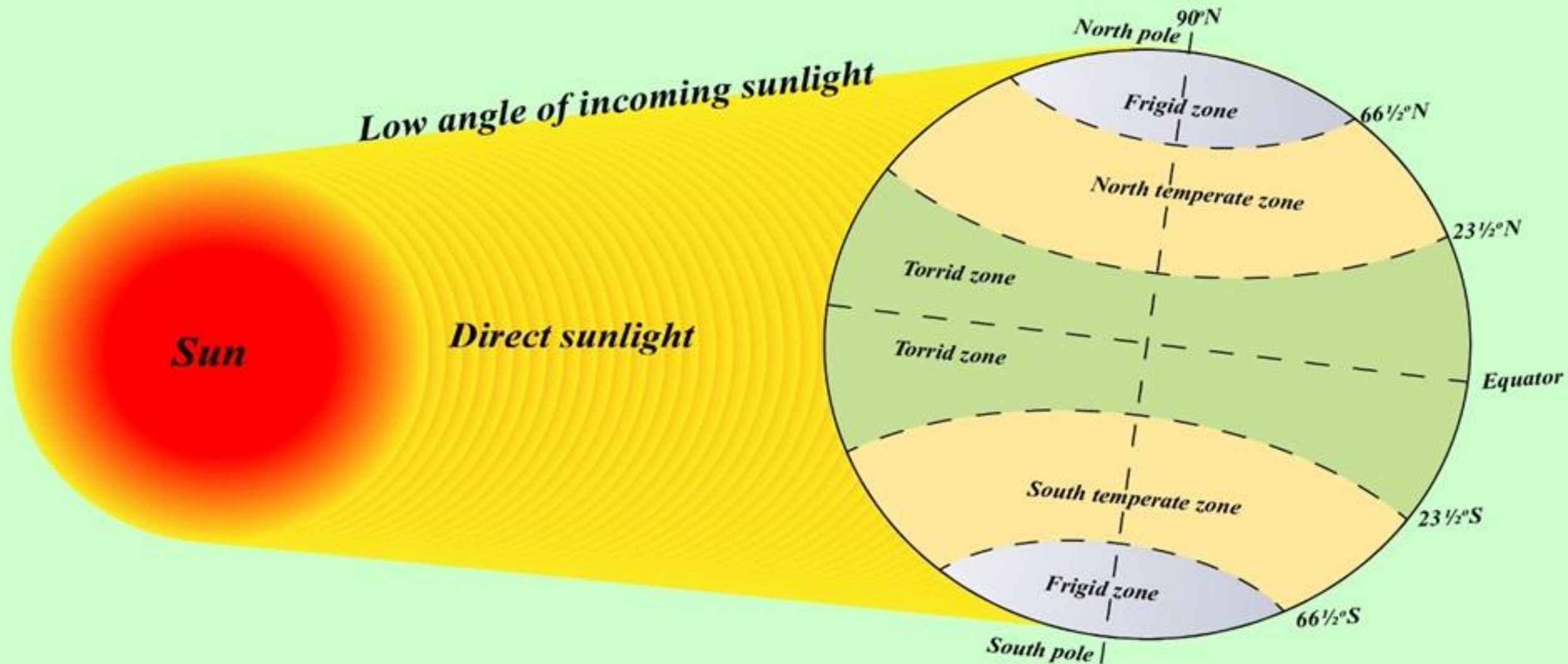
- The angle of the Sun's rays goes on decreasing towards the poles.
- Hence the regions between Tropic of Cancer and the Arctic Circle as well as Tropic of Capricorn and Antarctic Circle have moderate temperature.
- These are called temperate zones.

3. FRIGID ZONE

- ✓ The areas lying between Arctic Circle and North Pole as well as Antarctic Circle and South Pole are very cold.
- ✓ It is because here the Sun does not raise much above the horizon.
- ✓ Its rays are always slant and provides less heat.
- ✓ Hence these are called frigid (very cold) zones.



HEAT ZONES OF THE EARTH



WHAT IS THE RESULT OF THE UNEVEN HEATING OF THE EARTH?

THE EMERGENCE OF WINDS

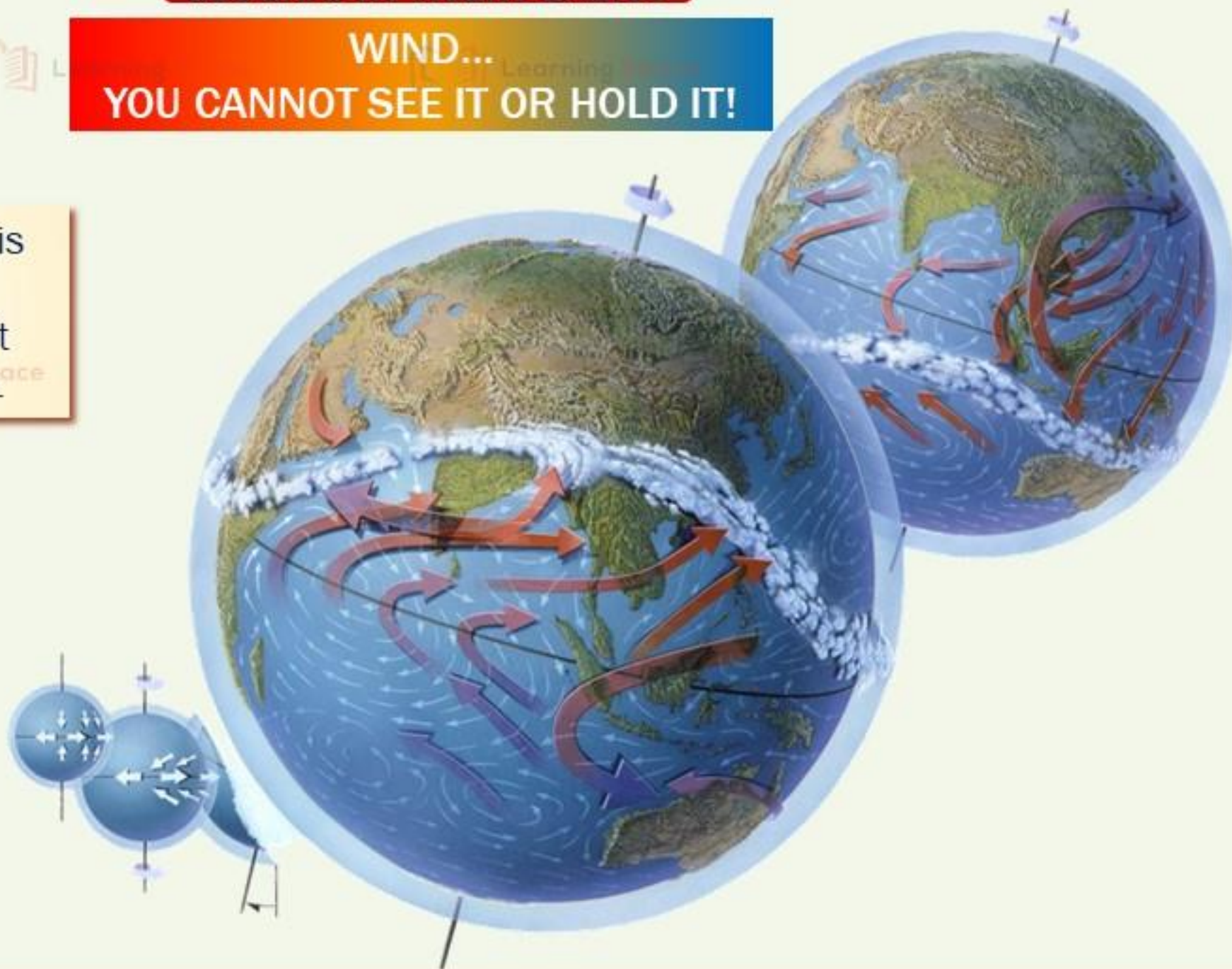
- Due to uneven heating of the Earth by the Sun as discussed, air moves from one place to another place.
- Differential heating results in differences in the atmospheric pressures.
- Due to the differences in the atmospheric pressure, air moves from higher to lower pressure area.
- It results in winds of various speeds and types.





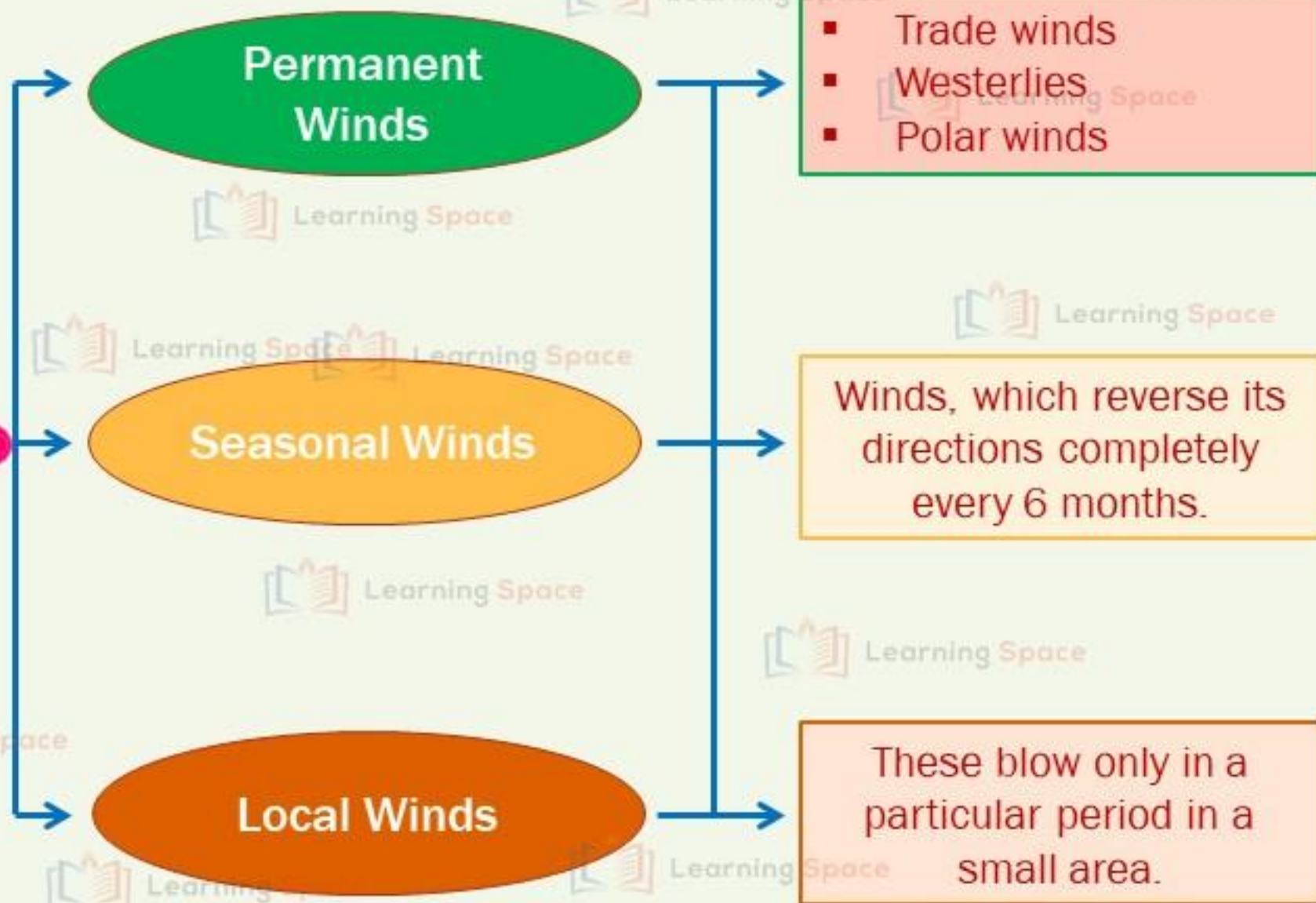
WIND... YOU CANNOT SEE IT OR HOLD IT!

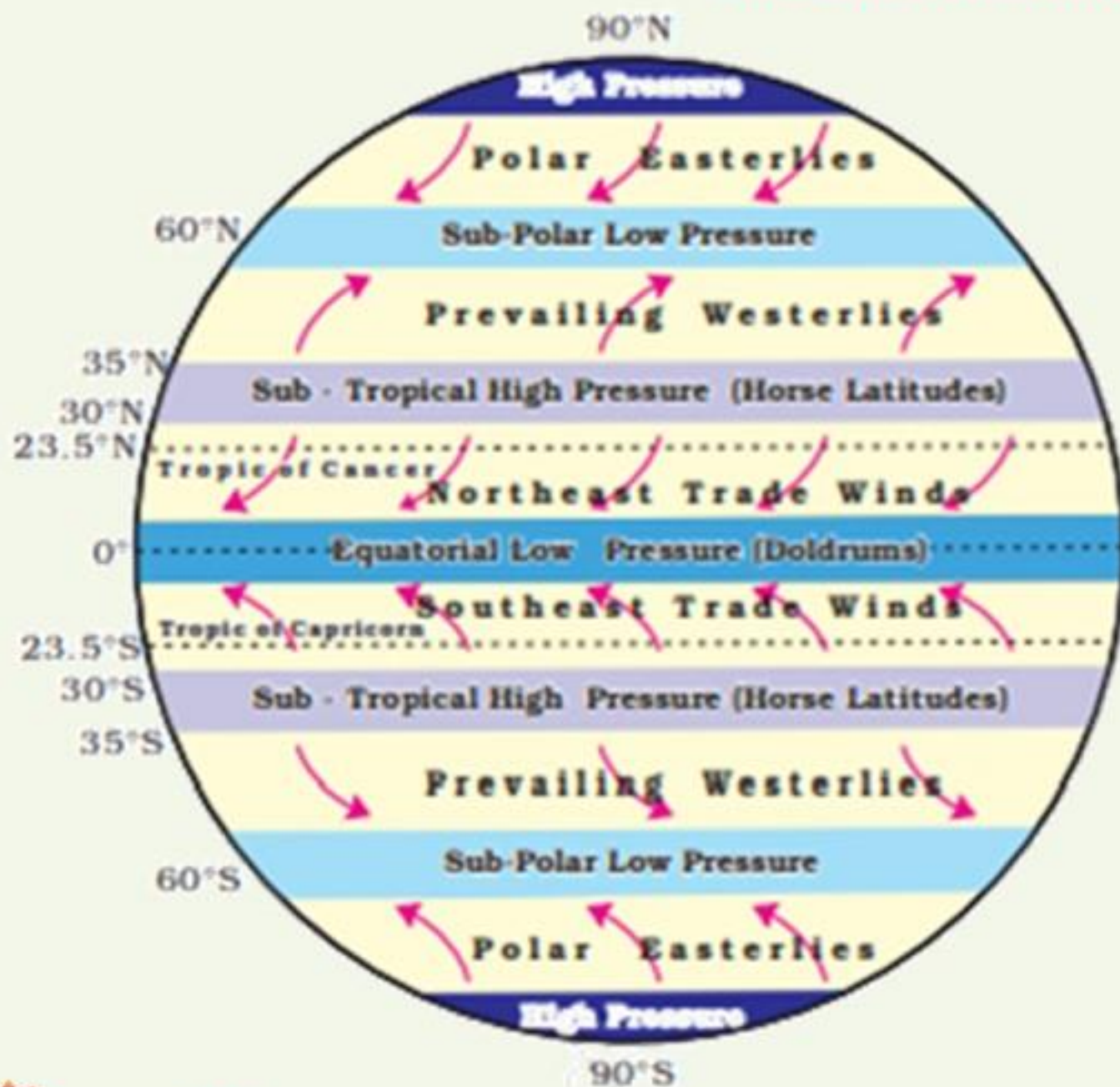
Atmospheric Pressure: It is the force exerted on a surface by the air above it as gravity pulls it to Earth.





CLASSIFICATION OF WINDS





Prevailing winds - winds that blow from a single direction over a specific area of the Earth.



PERMANENT WINDS

1. TRADE WINDS

- These are the powerful prevailing winds that blow from the east across the tropics.
- They have been instrumental in the history of exploration, communication, and trade.
- Strong trade winds are associated with a lack of precipitation, while weak trade winds carry rainfall far inland.
- Most tropical storms, including hurricanes, cyclones, and typhoons, develop as trade winds.

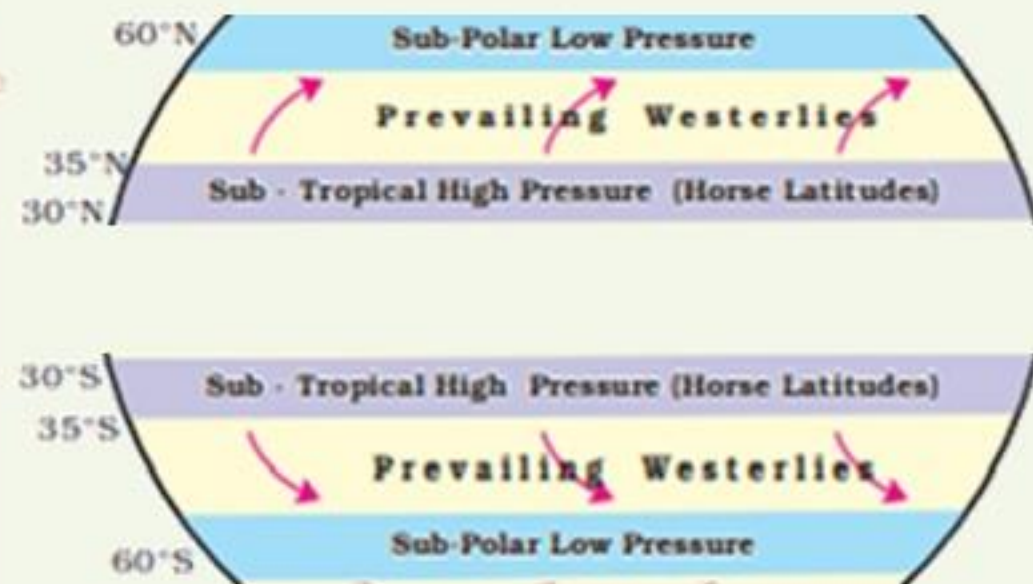
Precipitation: All forms in which water falls to Earth from the atmosphere.



2. WESTERLIES

- ✓ The prevailing winds that blow from the west at mid-latitudes i.e., the winds that blow between 30° and 60° Latitudes through out the year are known as westerlies.
- ✓ They are fed by polar winds and winds from the high-pressure horse latitudes.
- ✓ *Westerlies are strongest in the winter, when pressure over the pole is low*
- ✓ The strongest westerlies blow through the "Roaring Forties".

Roaring Forties - A wind zone between 40° and 50° latitudes.

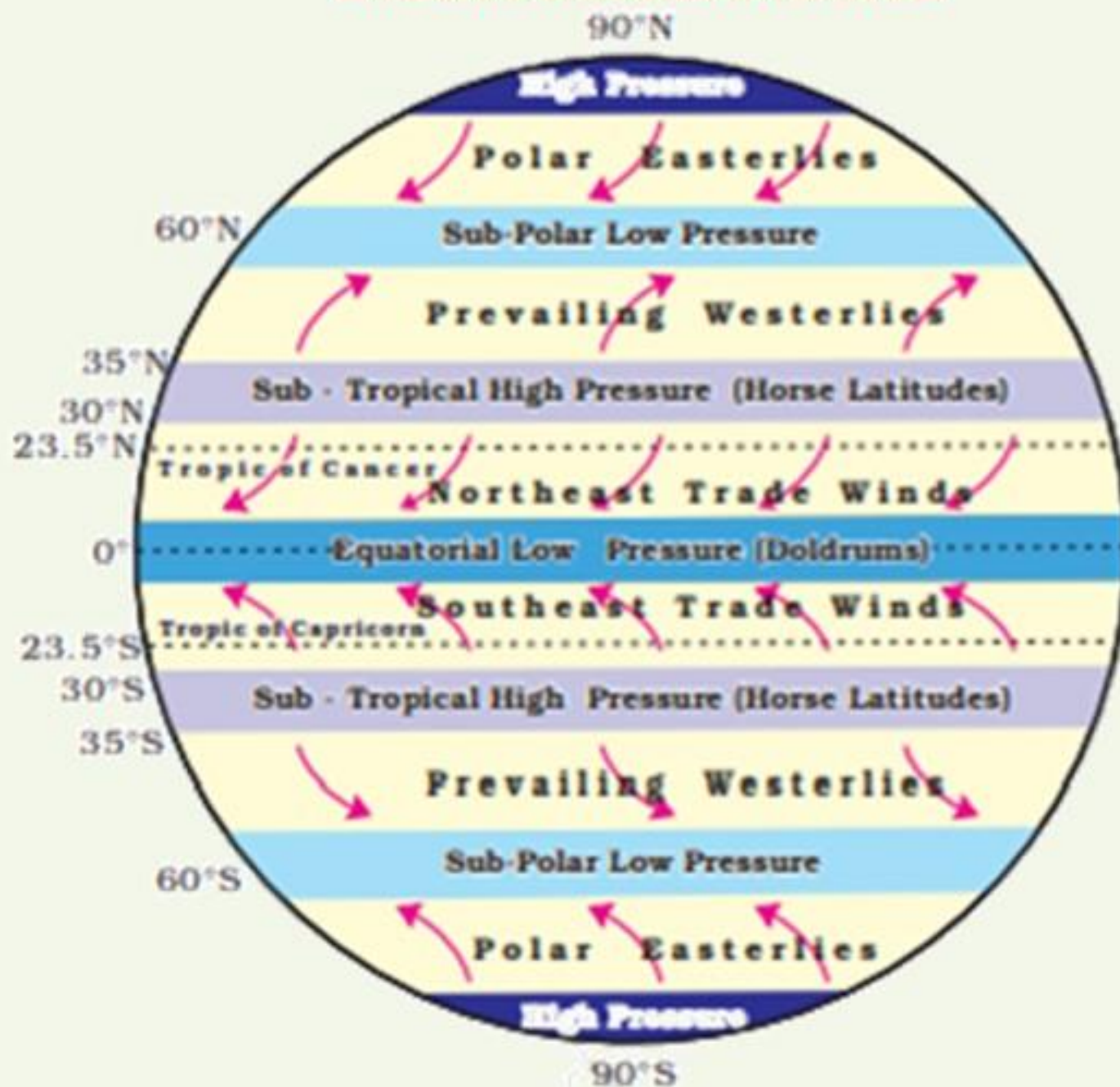


PERMANENT WINDS

3. POLAR WINDS

- Prevailing winds in the polar regions are Easterlies.
- Polar easterlies are dry, cold prevailing winds that blow from the east.
- They emanate from the polar highs, areas of high pressure around the North and South Poles.







SIMPLE PHENOMENA FOR CLARITY...

Wind: It is the horizontal movement of air, caused by the uneven heating of the Earth by the sun and the Earth's own rotation.

DID YOU KNOW!

Air: Layer of gases surrounding Earth.

- ✓ At the equator warming of land and water is higher in comparison to the rest of the globe.
- ✓ Air gets hot and raises higher to the atmosphere.
- ✓ The region around equator is filled by cooler denser air.
- ✓ This cyclic phenomena causes thunder storms, cyclones, and several complex weather systems!

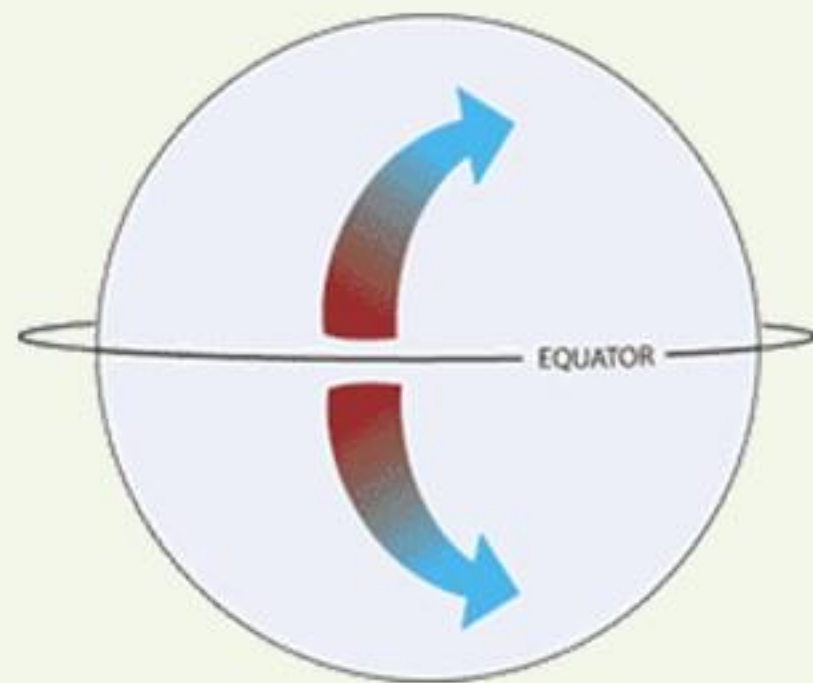


WHY THE WINDS MOVE EAST AND WEST RATHER THAN NORTH AND SOUTH?

This is because of Earth's rotation which is from West to East which generates what is known as the **Coriolis effect**.

CORIOLIS EFFECT

- Coriolis effect is the phenomenon that causes fluids like water and air to curve as they travel across or above earth's surface.
- As the earth is spherical the points on the equator move faster when compared to the location nearer to the poles.
- Because of the differential speeds of the locations at the equator and to the locations at the poles, if the wind wants to move straight, it cannot. It will get deflected.





TROPIC OF CANCER PASSES THROUGH EIGHT ASIAN COUNTRIES, WHICH INCLUDES TAIWAN

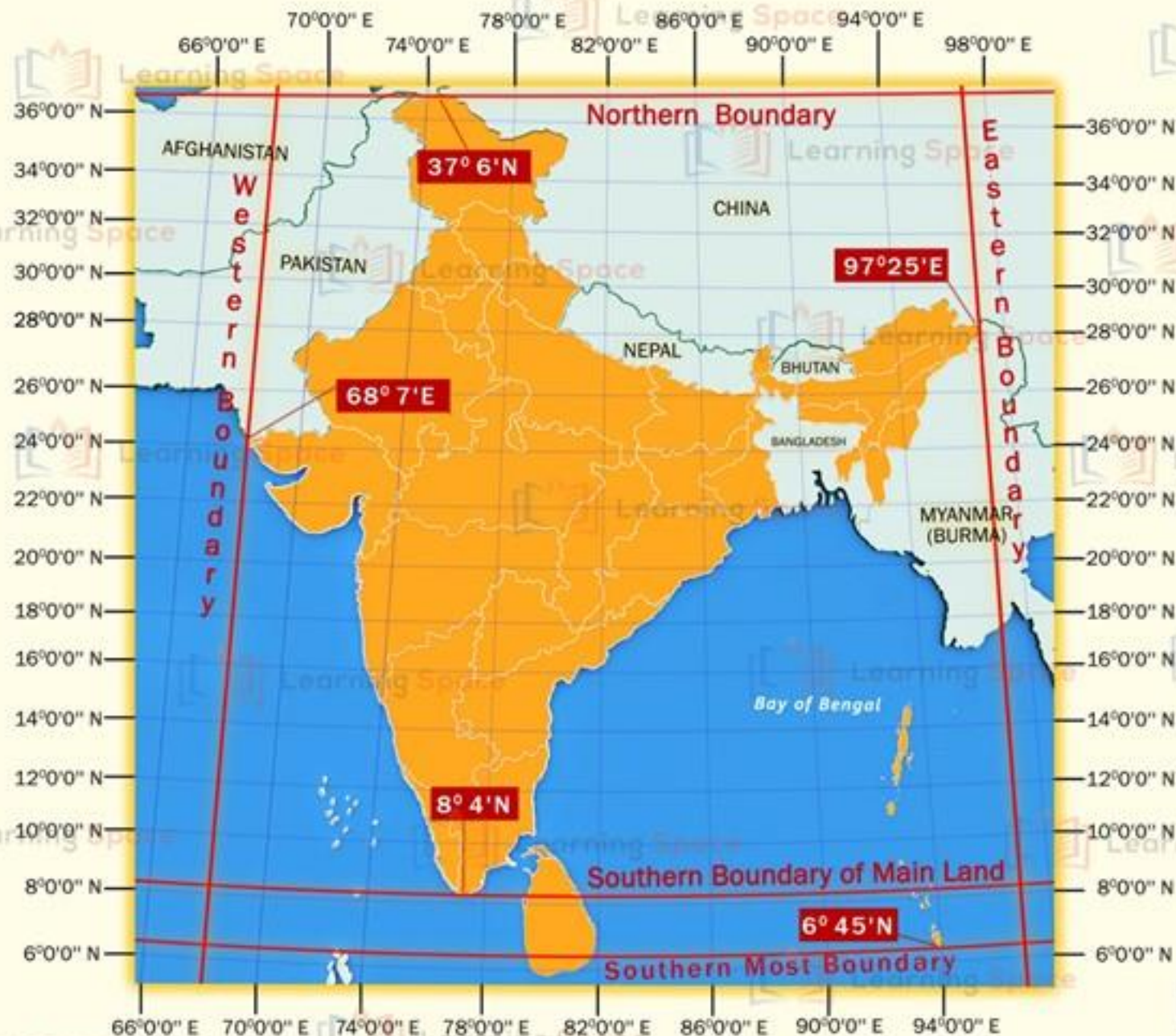




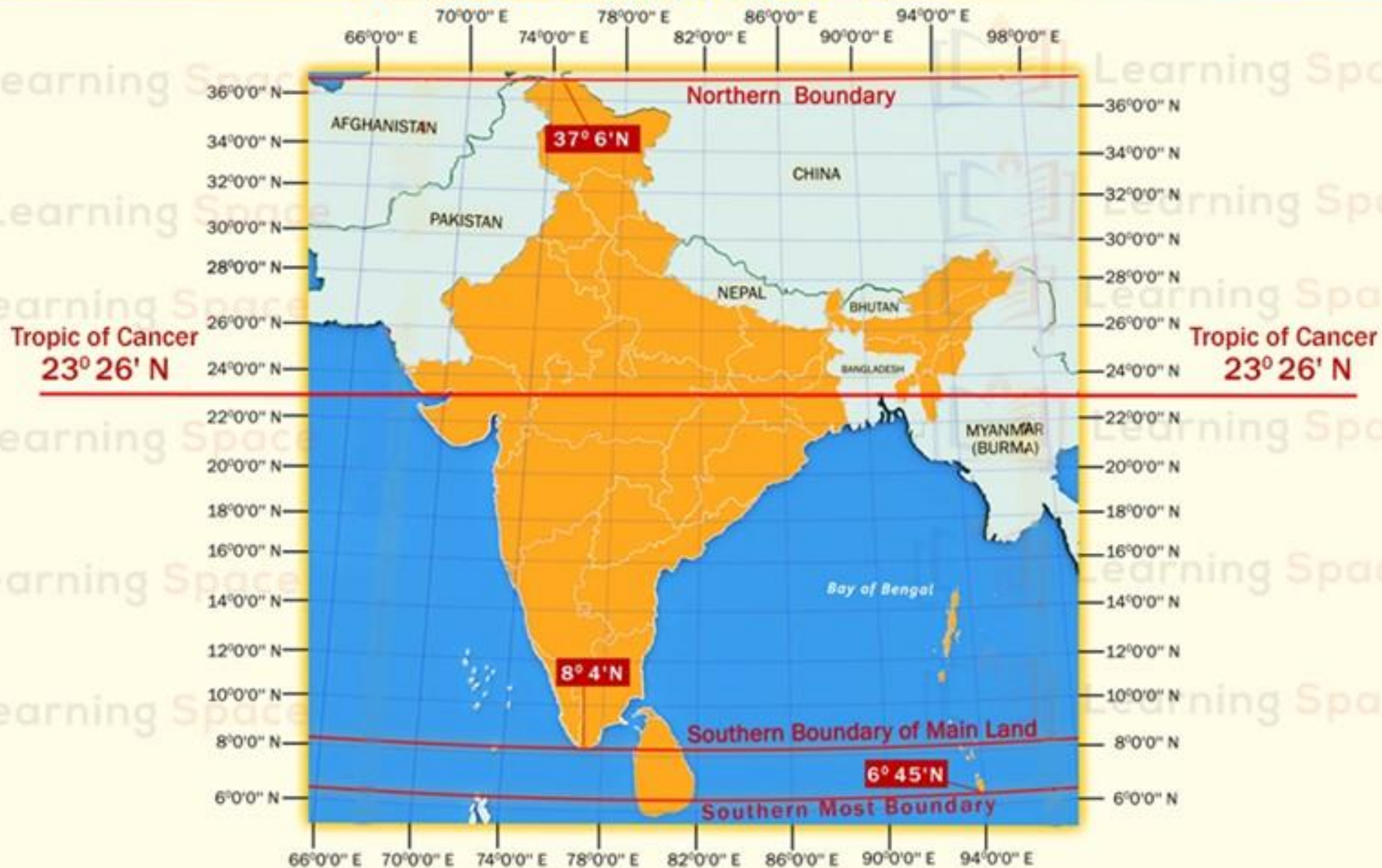
EQUATOR PASSES THROUGH TWO ASIAN COUNTRIES



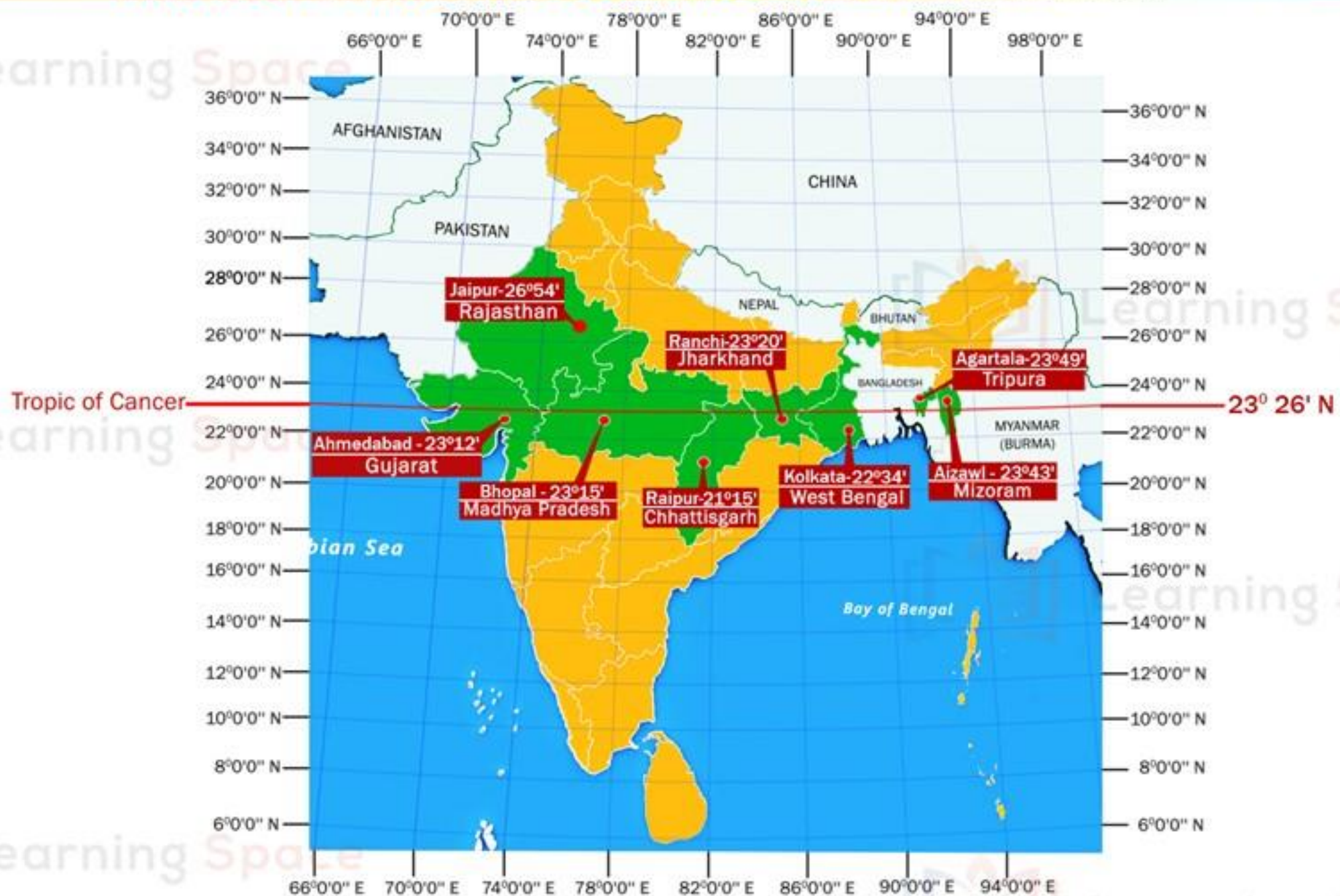
INDIA THROUGH PARALLELS AND MERIDIANS



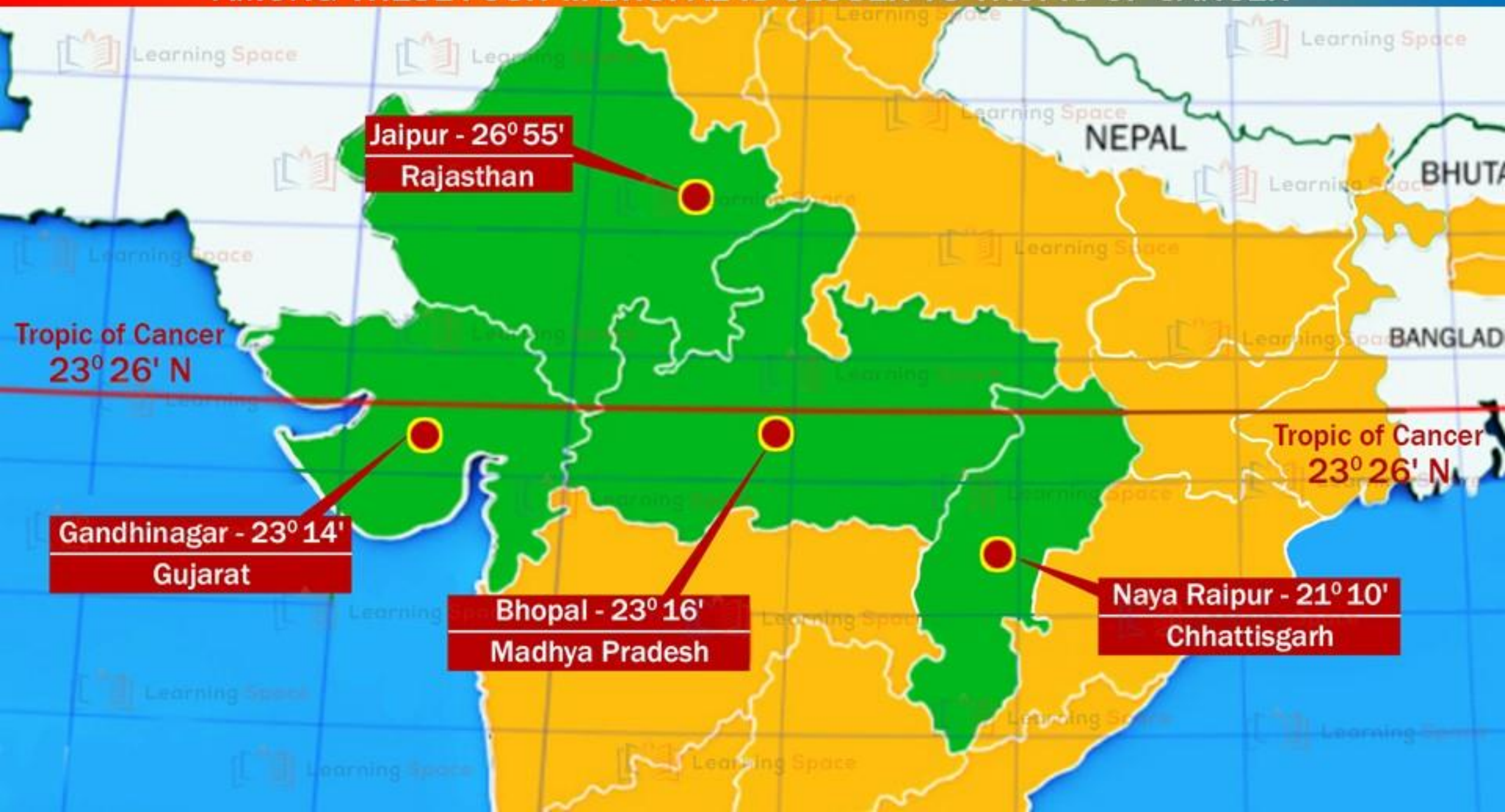
PARALLELS THROUGH INDIA



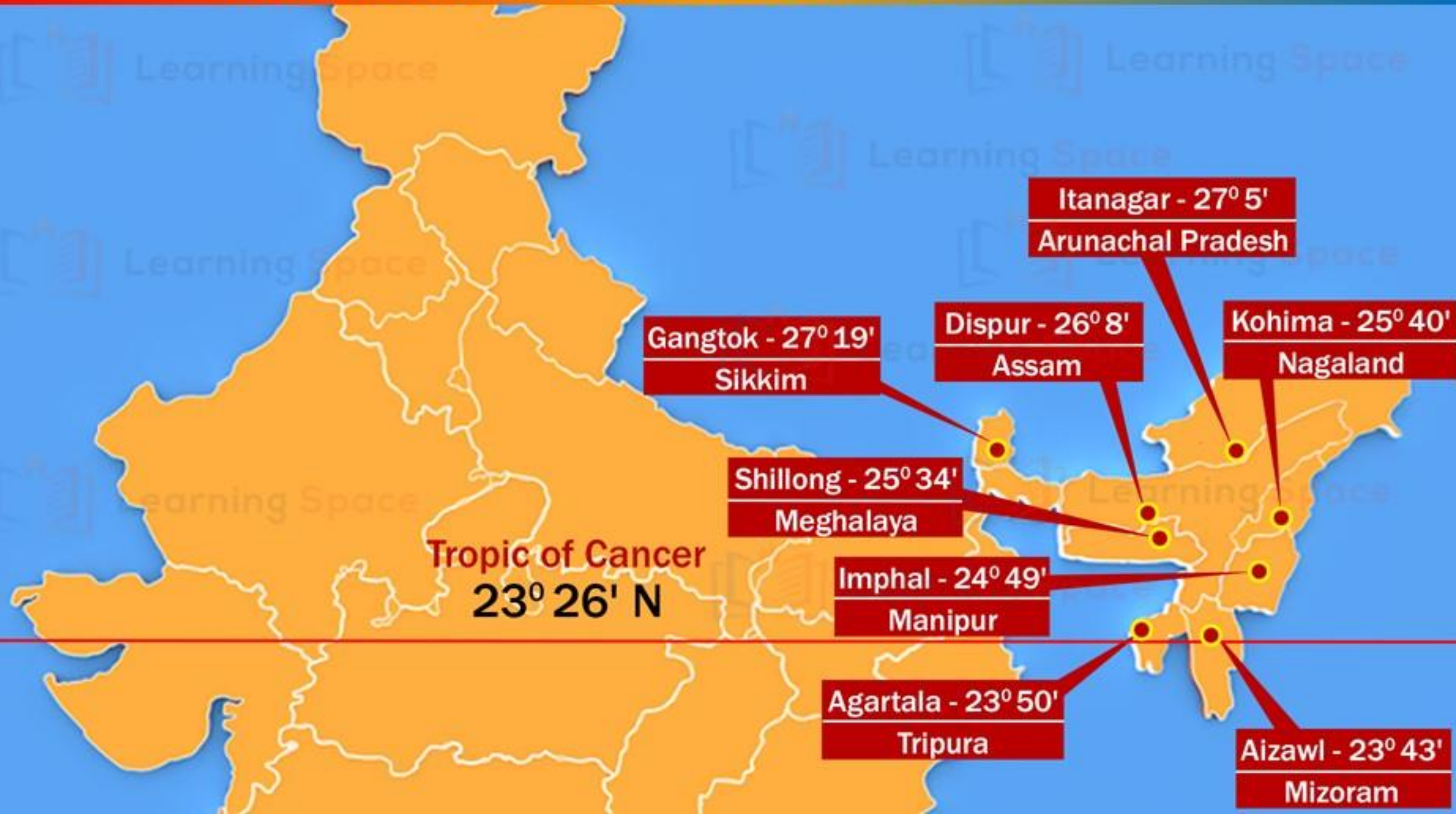
TROPIC OF CANCER PASSES THROUGH 8 STATES IN INDIA



AMONG THESE FOUR ... BHOPAL IS CLOSER TO TROPIC OF CANCER

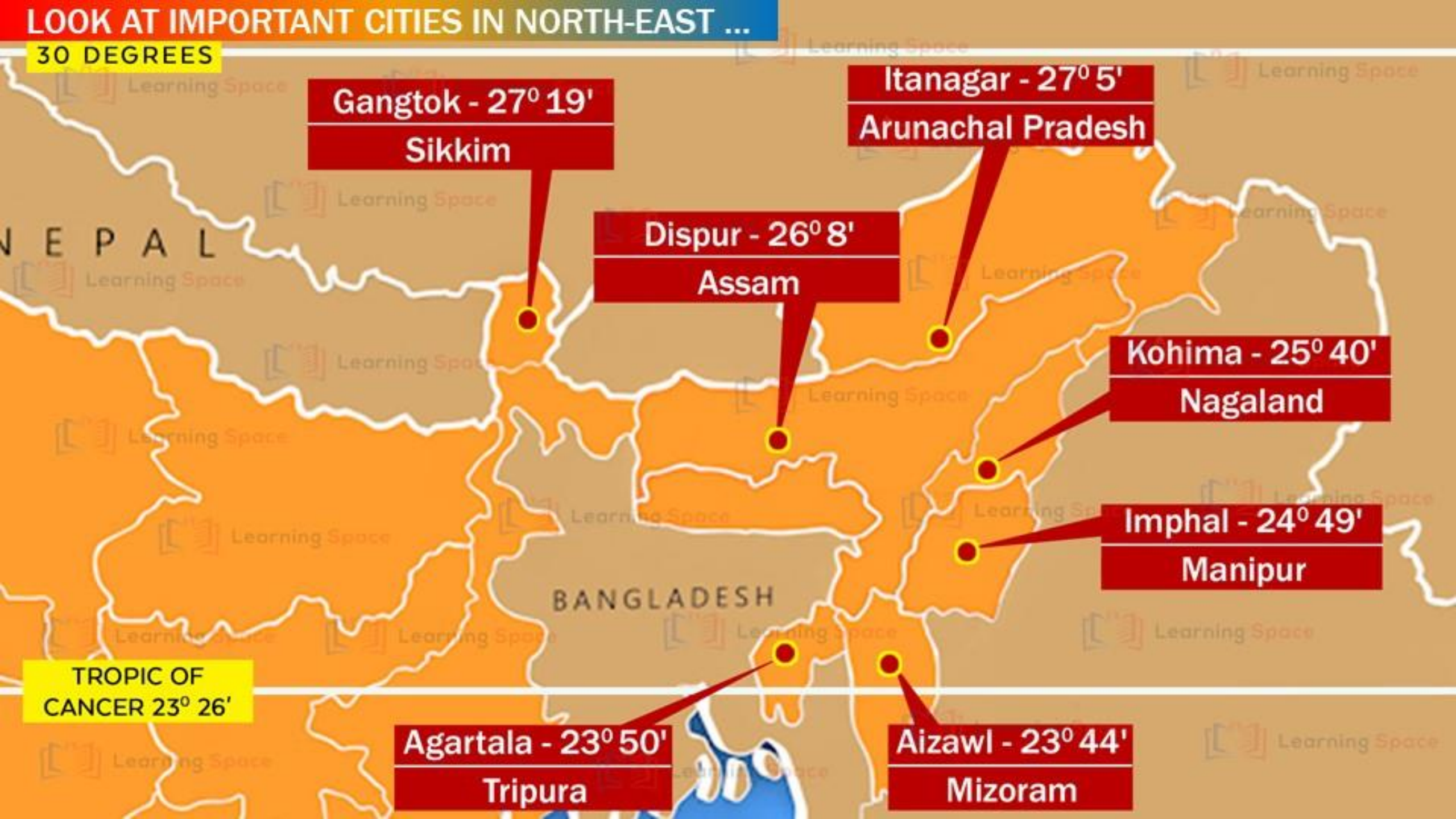


HAVE A GLANCE AT THE NORTH EASTERN STATES OF INDIA!



LOOK AT IMPORTANT CITIES IN NORTH-EAST ...

30 DEGREES



Gangtok - 27° 19'

Sikkim

Itanagar - 27° 5'

Arunachal Pradesh

Dispur - 26° 8'

Assam

Kohima - 25° 40'

Nagaland

Imphal - 24° 49'

Manipur

TROPIC OF
CANCER 23° 26'

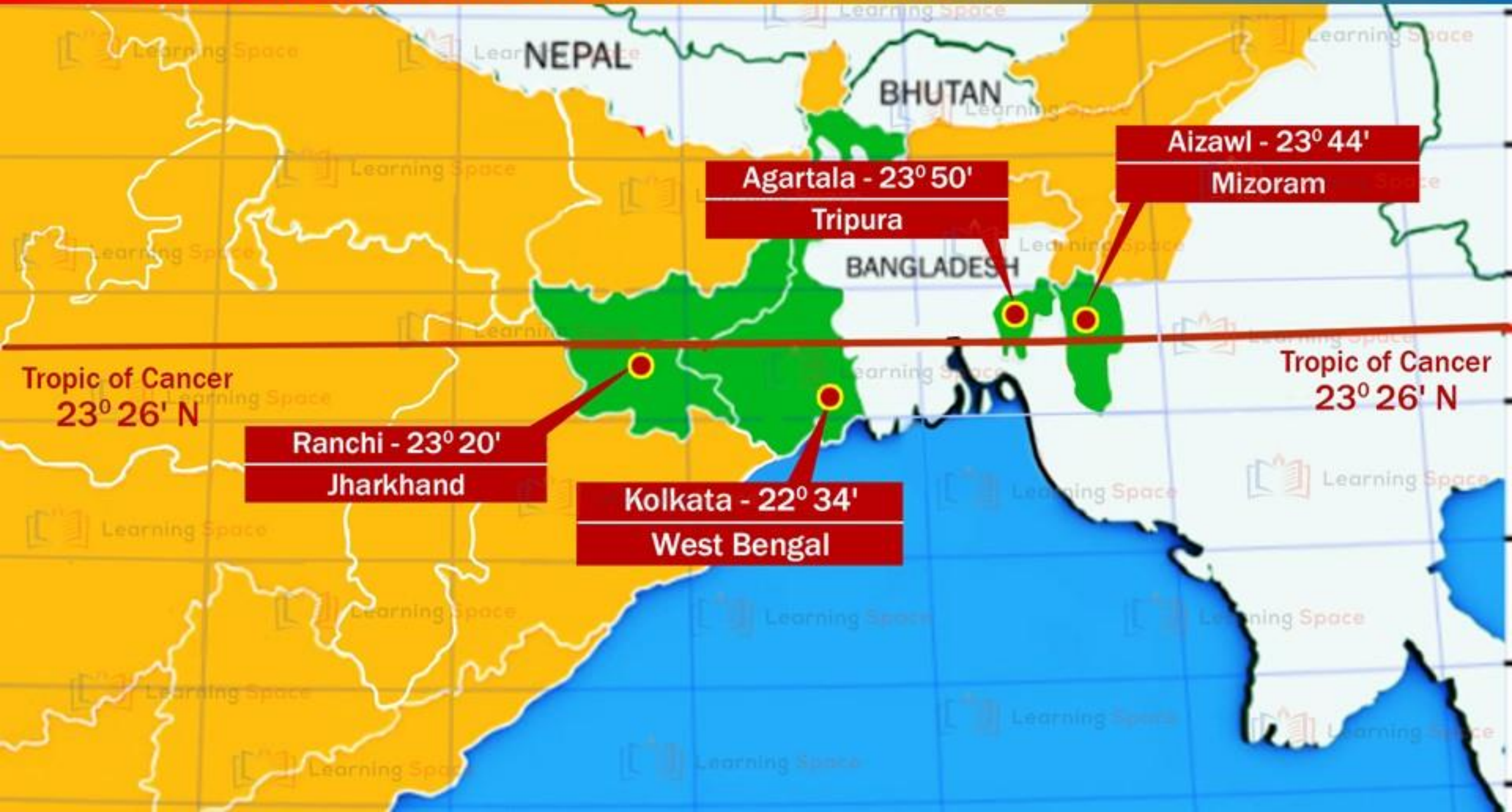
Agartala - 23° 50'

Tripura

Aizawl - 23° 44'

Mizoram

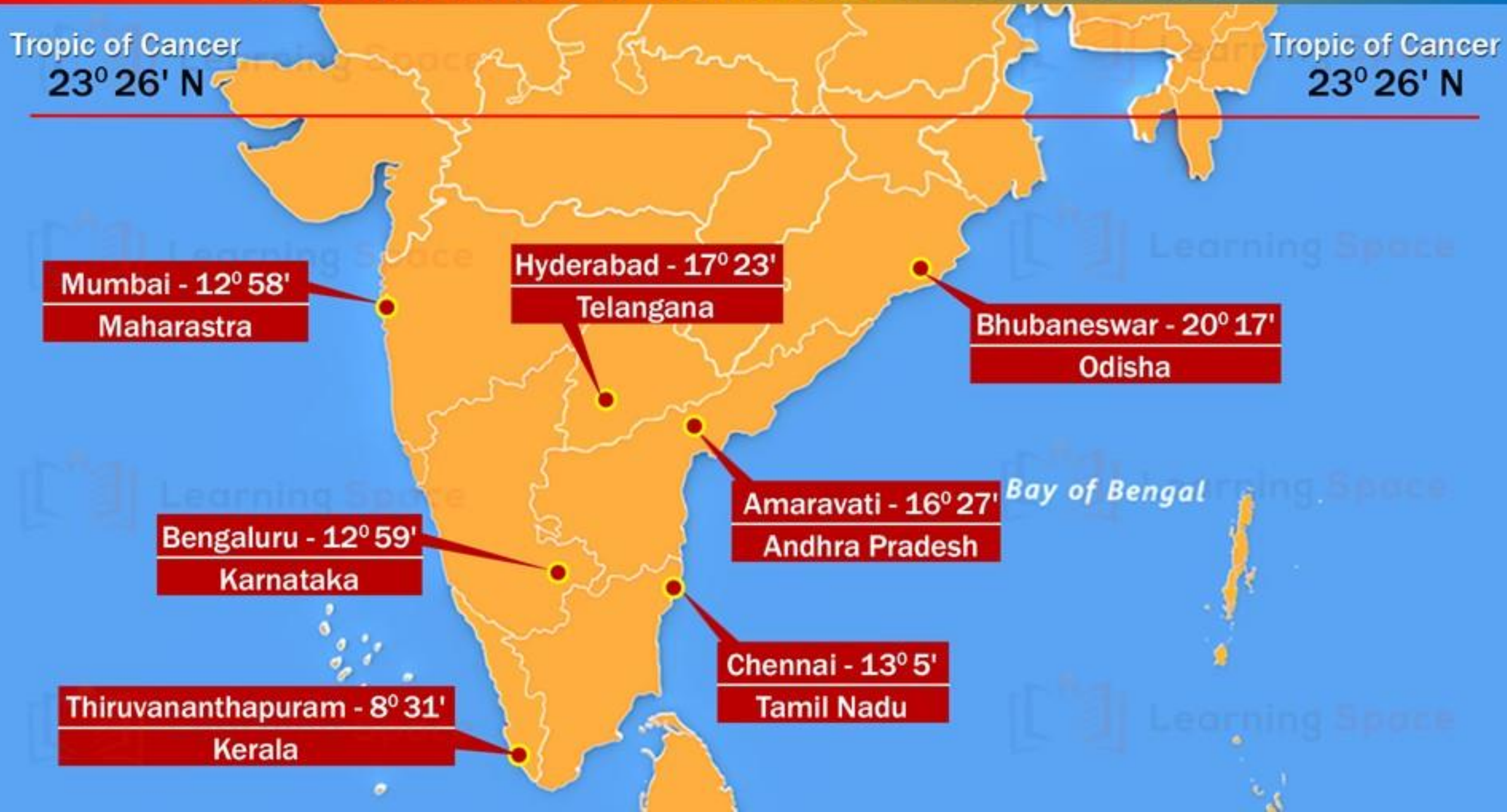
AMONG ALL THE EIGHT STATES ... RANCHI IS CLOSER TO TROPIC OF CANCER!



LOOK AT THE NORTHERN-MOST PART BEYOND 30 DEGREE!



KNOW THE LOCATON OF THE CAPITALS OF SOUTHERN STATES!



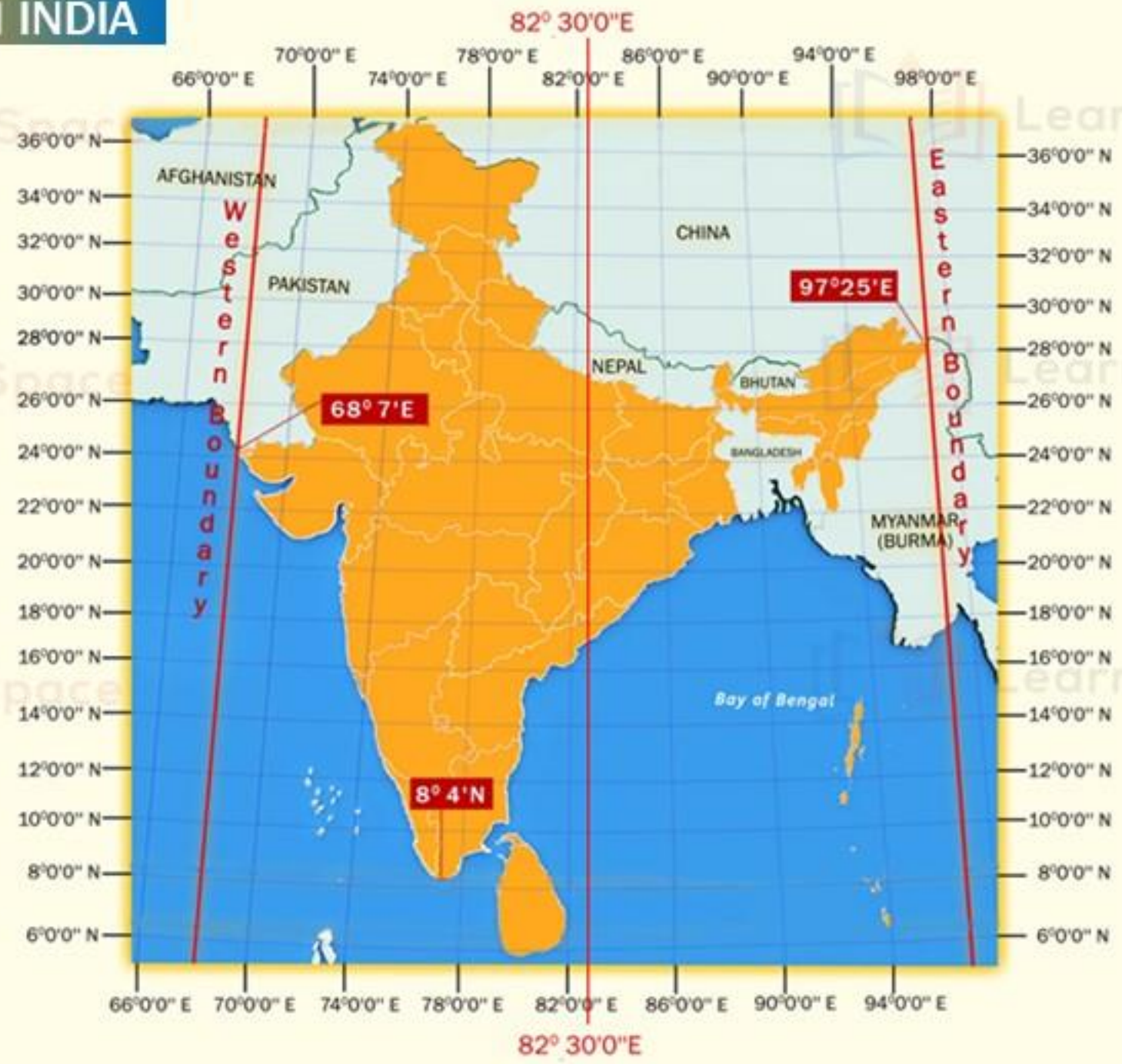
THE 10 DEGREE CHANNEL



WHY THERE IS TIME DIFFERENCE FROM ONE PLACE TO OTHER PLACE?



MERIDIANS THROUGH INDIA



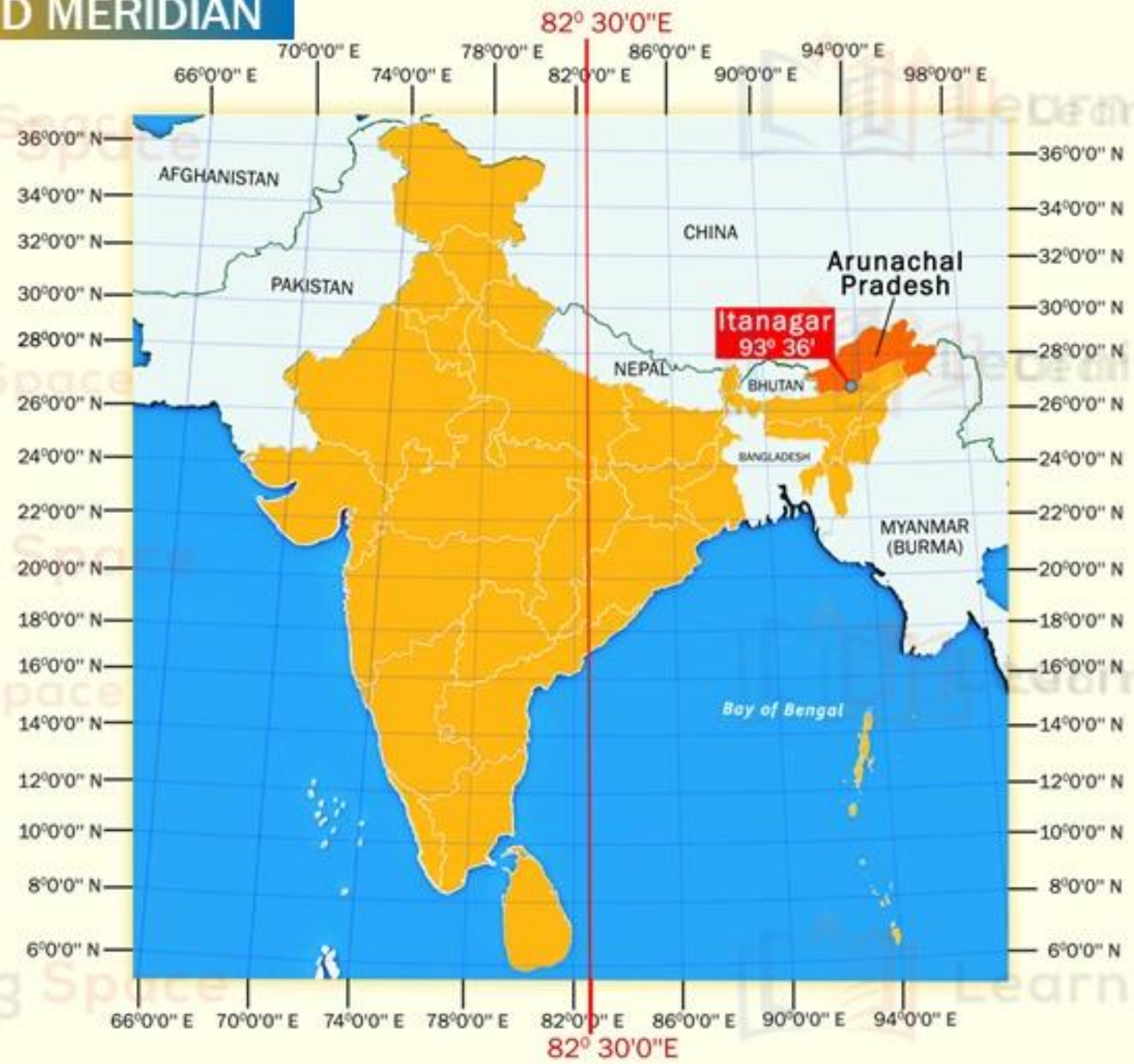


WHY THERE IS TIME DIFFERENCE FROM ONE PLACE TO OTHER PLACE?

- The Earth rotates from West to East.
- Hence, the places to East of the standard meridian will be ahead of the places to the West of the same.
- Roughly speaking the Earth rotates 360 degrees in about 24 hours. So, its almost 15 degrees in an hour or 1 degree in 4 minutes.
- Hence the time difference for each minute in difference of Longitude is 4 minutes.
- There's almost a difference of 30 degrees between the Western most part in Gujarat to the Eastern most part in Arunachal Pradesh. Hence the sun will raise almost two hours later in Gujarat.



THE INDIAN STANDARD MERIDIAN





WHY IS 82.5° E LONGITUDE TAKEN AS THE STANDARD MERIDIAN?

- As discussed earlier, there's a time lag of two hours between Arunachal Pradesh and Gujarat.
- As we move from Gujarat towards Arunachal Pradesh, the sun rises earlier similarly there are changes in the duration of days and night as we go from Kanyakumari to Kashmir!
- Hence some middle ground has to be found which can act as a reference line.
- Like for India, 82.5° E Longitude is taken as the Standard Meridian which passes through Mirzapur in Uttar Pradesh.



**LOOK AT THE LOCATION OF FIVE
CAPITALS OF THE STATES THROUGH
WHICH STANDARD MERIDIAN PASSES!**



LOCATION OF MAJOR CITIES CLOSE TO THE 82.5° E MERIDIAN



**WESTERN-MOST CAPITALS ARE
GANDHINAGAR & KAVARATTI!**

68° 7' E

74° E

Gandhinagar - 72° 38'

Gujarat

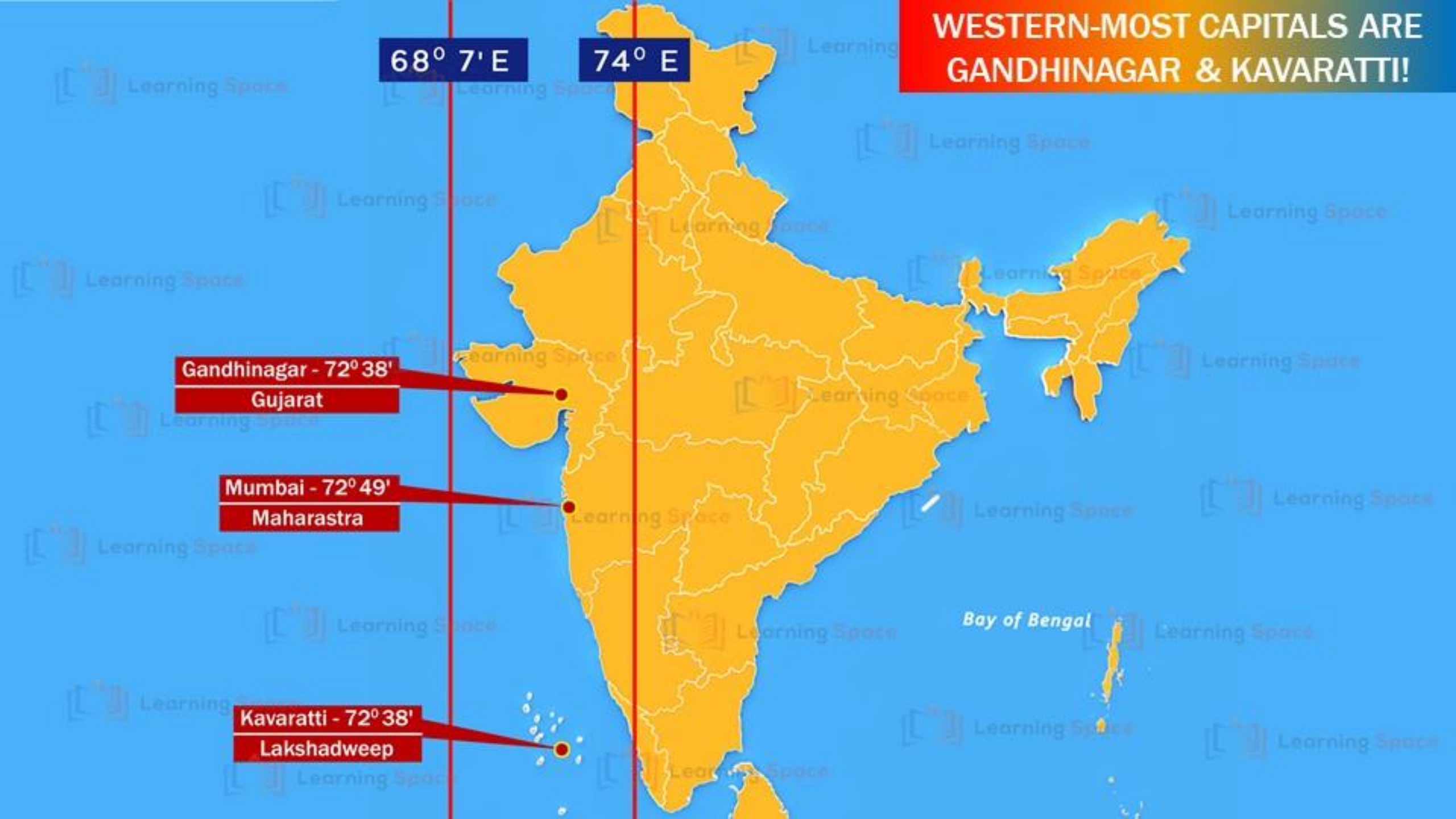
Mumbai - 72° 49'

Maharashtra

Kavaratti - 72° 38'

Lakshadweep

Bay of Bengal



KOHIMA IS THE EASTERN-MOST CAPITAL IN INDIA!



82° 30' E

97° 25' E

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

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